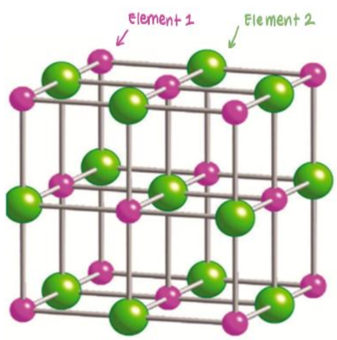
Station 1 – Building A Mineral (18 points)

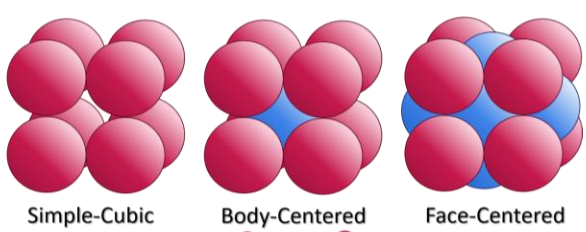
1. To build a mineral, we need to be able to understand what exactly a mineral is. On your answer sheet, list four characteristics an object needs to meet before we can formally declare it a mineral. (2)
2. Given a list of 5 objects, classify them as either a mineral or not a mineral. (5)
   a) Opal
   b) Ice
   c) Sulfur
   d) Gold
   e) Amber
3. One of the minerals on the list is halite, which has a cubic crystal habit. The following parts of question 3 guide you through to understanding halite's cubic crystalline structure.
   a) What is the chemical formula for halite? (1)
   b) Examine the diagram below and answer the following questions.



      i) Based on the diagram, how many Cl- ions is Na+ bonded with? (Make sure to visualize it in 3D for the correct answer!) (1)
      ii) What is name of the polyhedral shape based on the arrangement of Na+ and Cl-? (think: tetrahedral, cubic, linear… etc) (1)
   c) Now think about the reverse: what is the coordination number between Na+ and Cl-? What information does this tell us about the bonding structure of Na and Cl? (2)

   Crystals are made up of smaller objects referred to as "unit cells". There are three main types of unit cells: primitive, body-centered-cubic (BCC), and face-centered cubic (FCC), which build up to make a larger structure referred to as a crystal lattice by the same name.

   d)



      i) Based on the diagram, which type of unit cell best represents halite? (1)
      ii) Using the diagram, count how many Cl- atoms are fully owned by a singular unit cell. Do the same for Na+. Show your work! (4)
4. Question 3 illustrates a key concept that is important in understanding crystal lattices. What is the name of the law that states that the angles within the various faces of a crystal remain unchanged throughout its growth called? (1)

<u>Station 2 (16 points)</u>


A


B


C


D


E


F

1. Identify the minerals A-F. (6)
2. The following questions will be about the chemical composition of the minerals above.
    a. Which 2 specimens have the same chemical composition? (2)
    b. What term is used to describe elements that can take on 2 or more different physical forms? (1)
    c. What difference in bonding structure makes these 2 specimens so different in morphology? (2)
3. The following questions will be about ores.
    a. Which three specimens have been used as coinage metals? (1)
    b. What chemical property allows these minerals to exist in their pure elemental form?(1)
        a) Low reactivity
        b) Electronegativity
        c) Stable Crystal Structure
        d) Diaphaneity
    c. Out of the three, which is commonly found in households for wiring? (1)
    d. What mineral on the list is confused with specimen A for its similarity in color and as a result, has a nickname that refers to its trickery? (1)
4. While other minerals have different color variations, Specimen F is known for its bright lemon-yellow color and can be easily identified by this distinction. How do electrons contribute to Specimen F's color? (1)
    a) Their orbitals correspond to different wavelengths on the electromagnetic spectrum
    b) They change the hardness and the specific gravity of the mineral
    c) They absorb all colors and reflect yellow
    d) Its birefringence and symmetry contribute to its yellow color

Station 3 (8 points)

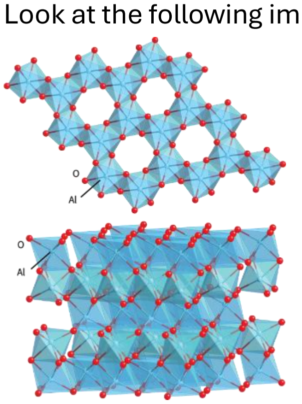
   

A B C D

E F

1. The term describing all 6 of these specimens is known as what? (1)
2. Two of the specimens at this station are the same mineral, just in different forms. What specimens are they? (2)
3. What element gives Specimen A its unique pink/red color? (1)
    a) Aluminum
    b) Chromium
    c) Magnesium
    d) Iron
4. Specimen C is known to have other specimens with the same color, but not the same quality. What chemical difference makes Specimen C a deep green compared to pale green specimens? Answer with the chemical element's name and NOT the chemical symbol. (1)
5. Most of these minerals are known for their hardness on the Mohs scale. The following parts will help you explain what makes these minerals so high on the scale.
    a. What is the hardness of Specimen B? (The answer is a whole number) (1)
    b. Look at the following image:

    This image is an image of a structure in Specimen B referred to as close packing. There are two types of close packing for minerals: cubic and hexagonal close packing. Based on what you know about Specimen B, what kind of close packing do you think is shown in the diagram? (1)
    c. However, when you look at the diagram, it does not look like it is a true closely packed structure due to the octahedral sites being occupied by ________________ ions. (1)

Station 4 (18 points)

1. The following questions will be able fracture and cleavage.
   a) Define fracture and cleavage on your answer sheet. (2)
   b) Given a set of minerals, circle either poor, fair, good/perfect, or no cleavage. (5)
      i) Mica
      ii) Talc
      iii) Diamond
      iv) Halite
      v) Quartz

   c) Which of the minerals in part B has no cleavage? In its atomic structure, what makes this mineral not be able to form any cleavage planes? (2)
   d) Cleavage is often determined by the atomic structure of the mineral. For example, when you break halite, you find perfect cubes after breaking it. Given this information, at what angle are the planes of weakness oriented in? (1)

2. The following questions will be about more mineral properties.
   a) Define luster and streak. (2)

   b) If I were to perform a streak test on a mineral with a hardness greater than 7, will it produce a streak? Why or why not? (2)

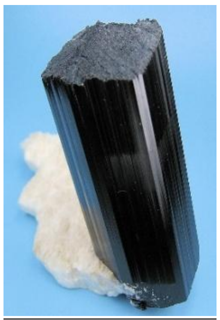
   c) Bob finds an unknown mineral and uses apatite to scratch the surface of the mineral and finds out that it scratches the surface of the mineral. He then uses feldspar to scratch the surface and finds that it does not scratch the surface of the mineral.
      The hardness of the unknown mineral is between _____ and _____. (1)

3. Specimens of orthoclase have perfect cleavage on {001} and a good cleavage on {010}. Mark the diagram where these planes occur. Make sure to label your axes and show how you were able to determine which faces correspond to the Miller indices. (3)

Station 5 (17 points)

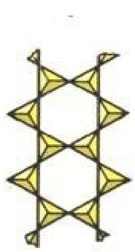
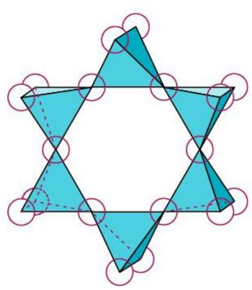
  

A B C

D E F

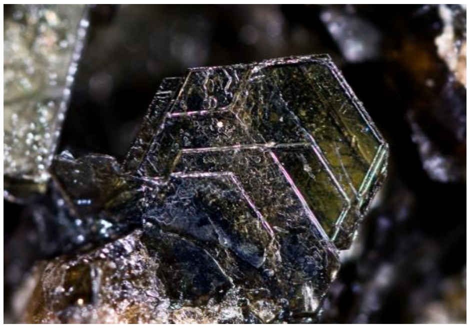
1. Identify the specimens. (6)
2. What is the general formula for these types of minerals? (1)
3. However, depending on the type of mineral, this formula will be altered. Why does the formula change depending on the group (or type of mineral)? (1)
4. The following questions will be about Specimen C.
   a) Specimen C is also known as a polymorph. What does this mean? (1)
   b) What environmental changes create Specimen C's polymorphs? (2)
5. Match the following structures to the minerals at this station. (3)

Structure 1 Structure 2 Structure 3

6. Specimen A makes up what rock that makes up the majority of Earth's upper mantle? (1)
7. Specimen E is often found in what types of igneous rocks? (1)
8. In comparison, specimen F can be found in what types of igneous rocks? (1)

Station 6 (12 points)

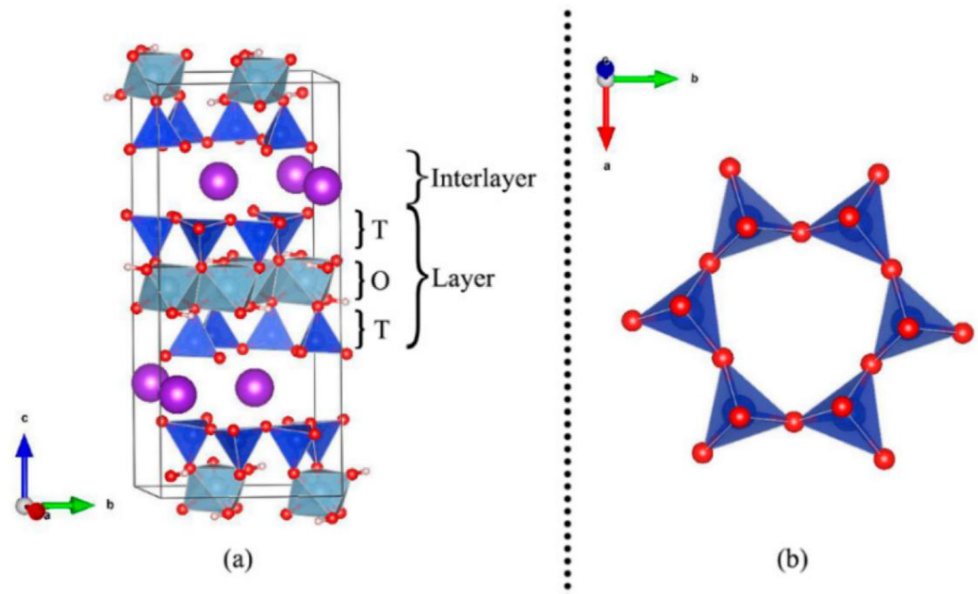


A

B

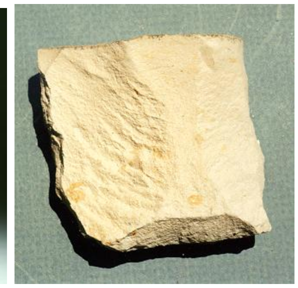
C

1. Identify the specimens at this station. (3)
2. Which of the specimens can be found in igneous rocks? (1)
3. Which of these specimens is used for K-Ar dating? (1)
4. Which of these specimens was used as a cheaper alternative to glass during the 16^{th} century? (1)
5. Specimen C has weak pleochromism. What does this mean? (1)
6. The following question will be split into parts regarding the diagram below.



a) Which of the specimens at this station most likely has this structure? (1)
b) This structure is referred to as dioctahedral. Using your knowledge of mineralogy, define what dioctahedral means. (2)
c) Which two sites make this mineral dioctahedral? (2)

Station 7 (15 points)

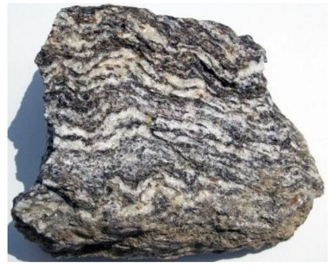
A B C D

1. Identify all the specimens at this station. (4)
2. What is the difference between a conglomerate and Specimen C? (2)
3. Select all of the following environments where a conglomerate can form: (3)
   a) River delta
   b) Aeolian environment
   c) Outwash plain
   d) Dry plain
   e) Beach
4. Which of the following specimens will most likely contain fossils? (1)
5. Which of the following specimens at this station would most likely have the structure shown below? (1)

6. Usually, these specimens will not have the structure shown in question 5. However, list one environmental change that will allow this structure to form. (2)
7. Which of the following specimens is the primary source of gallium? (1)
8. Which of the following environments can specimen D be found in? (1)
   a) Desert
   b) Continental shelf
   c) Swamp
   d) Beach

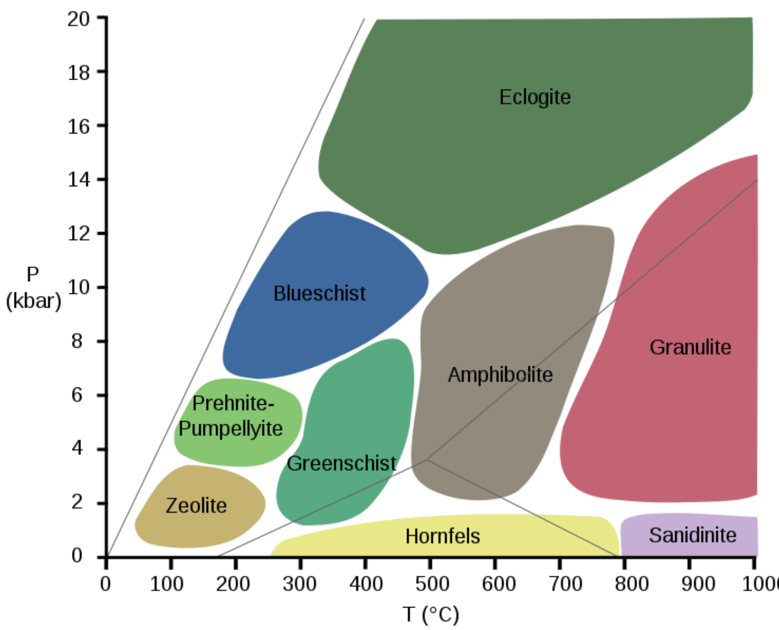
Station 8 (14 points)





A B C

1. Identify all the specimens at this station. (3)
2. Rank these specimens from the most foliated to the least foliated. (1)
3. Select all of the following conditions in which foliation is possible: (2)
   a) High heat, high pressure
   b) Low heat, low pressure
   c) Low heat, high pressure
   d) High heat, low pressure
4. Which of the following minerals could you expect in Specimen A? (1)
   a) Kyanite
   b) Muscovite
   c) Biotite
   d) Hornblende
5. What is the parent rock of Specimen B? (1)



6. The following questions will use the diagram above.

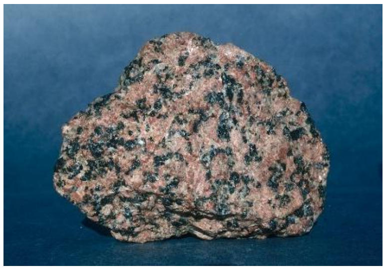
a) What is the name of this kind of diagram? (1)

b) Name the rock that will form at low pressure and high temperature. (1)

c) On your answer sheet, you have the same diagram. Mark the region where you believe that Barrovian metamorphism will occur. (2)

d) Mark the region on the diagram which rocks will be found at subduction zones. (2)

Station 9 (13 points)

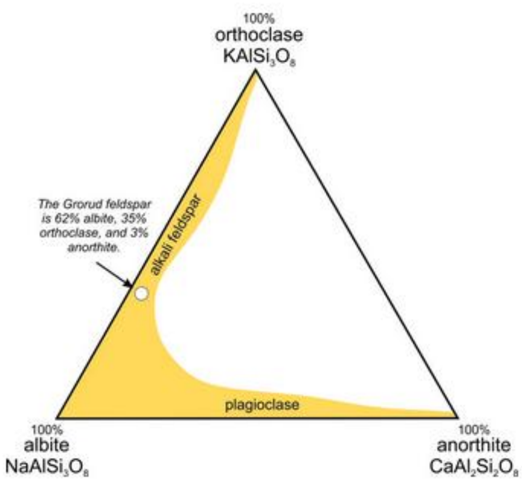

A


B


C

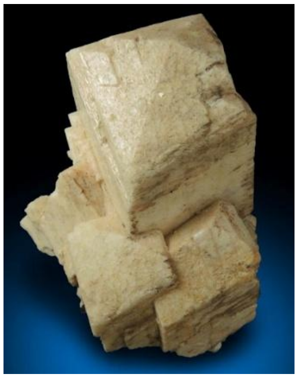

D

1. Identify the specimens at this station. (4)
2. Classify the specimens into intrusive and extrusive specimens. (2)
3. When Specimen A is pink colored, it indicates the presence of what mineral? (1)
4. Which of these make up the majority of continental crust? (1)
5. In what case would we use the TAS system instead of QAPF? (2)



6. Given the diagram above, answer the following questions.
   a) What is the name of this gap (aka the region of white space)? (1)
   b) Propose a reason why this gap might exist. (Hint: it has something to do with stability) (2)

Station 10 (13 points)



A

B

C

1. Identify the following specimens. (3)
2. Which of the following specimens at this station go through the process of twinning?(1)
3. Which of these specimens is most likely will form simple contact twins? (1)
4. Which of these specimens is most likely will form complex contact twins referred to as polysynthetic twins? (1)
5. The following question will pertain to Miller indices as well twinning.
    a) Miller indices are a notation for how we describe lattice planes in minerals. How might this relate to the process described in question 2? (1)
    b) Before we start working in a 3D plane, we'll work slowly and start drawing these indexes in 2D, where the z plane does not exist. Given the following indices, sketch a vector from the point marked {000} when you start drawing each vector. (3)
        i) [100]
        ii) [210]
        iii) [1-10]
    c) Let's try it in 3D. Specimen B is known for its twins on the {031} plane and the {231} plane. On your answer sheet there will be a diagram in which you will draw the {031} plane, provided with the axes. Make sure to show how you got there—hint: there is something that needs to be done with the numbers in order for the plane to be correctly plotted! (3)

Station 11 (12 points)

1) Select all of the following minerals that have the same crystal habit:
   a) Halite
   b) Pyrite
   c) Sphalerite
   d) Magnetite
   e) Garnet / Almandine
2) Which of the following environments would you NOT expect sandstone to form in?
   a) Desert
   b) Aeolian
   c) Deltas
   d) Deep Lake
3) Which of the following minerals is the primary ore of iron?
   a) Hematite
   b) Corundum
   c) Pyrite
   d) Stibnite
4) Which of the following minerals has the nickname "Peacock ore"?
   a) Willemite
   b) Bornite
   c) Copper
   d) Labradorite
5) If I find a piece of chalcedony, and I notice that it has black and white banding, it is referred to as what name?
   a) Moss agate
   b) Chalcedony
   c) Onyx
   d) Jasper
6) Which of the following minerals is NOT a nesosilicate?
   a) Augite
   b) Almandine
   c) Grossular
   d) Topaz
7) What is the difference between the discontinuous and the continuous branch of Bowen's Reaction series?
   a) Discontinuous branch shows the formation of feldspars while the continuous branch shows the formation of mafic minerals
   b) Continuous describes the evolution of compositions, while discontinuous represents distinct minerals
   c) Discontinuous branch shows the formation of metamorphic facies while the continuous branch shows the formation of igneous facies
   d) Discontinuous branch shows the formation of silicates, while the continuous branch shows the formation of oxides
8) Which of the following is a pair of pseudomorphs referred as to as a paramorph?
   a) Hematite and magnetite
   b) Corundum and pyrite
   c) Calcite and aragonite
   d) Turquoise and Amazonite
9) Given an unknown metamorphic facies, what rock would you expect to form under extremely high pressure and high heat?
   a) Serpentine
   b) Mica schist
   c) Blueschist
   d) Eclogite
10) If I had an unknown rock, but I knew that it went under high heat and pressure and transformed into a metamorphic rock, what type of rock would I expect my parent rock to be?
    a) Metamorphic
    b) Igneous
    c) Sedimentary
    d) Porphyritic
    e) Can not be determined
11) What kind of metamorphism is associated with convergent boundaries?
    a) Regional
    b) Contact
    c) Shear
    d) Dynamic
12) Which rocks would I most likely find at a faulting zone?
    a) Granite
    b) Shale
    c) Siltstone
    d) Rock Salt
    e) None of the Above

Station 12 - Speed ID (12 points)

A

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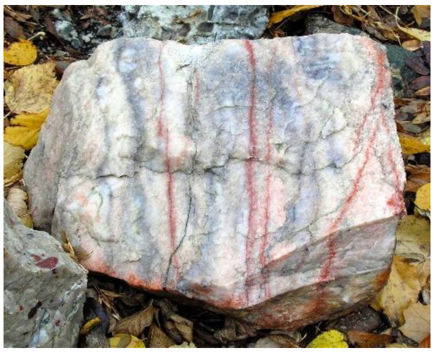

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